

16

CHAPTER

Options on Stock Indices and Currencies



Options on stock indices and currencies were introduced in Chapter 9. This chapter discusses them in more detail. It explains how they work and reviews some of the ways they can be used. In the second half of the chapter, the valuation results in Chapter 14 are extended to cover European options on a stock paying a known dividend yield. It is then argued that both stock indices and currencies are analogous to stocks paying dividend yields. This enables the results for options on a stock paying a dividend yield to be applied to these types of options as well.

16.1 OPTIONS ON STOCK INDICES

Several exchanges trade options on stock indices. Some of the indices track the movement of the market as a whole. Others are based on the performance of a particular sector (e.g., computer technology, oil and gas, transportation, or telecoms). Among the index options traded on the Chicago Board Options Exchange (CBOE) are American and European options on the S&P 100 (OEX and XEO), European options on the S&P 500 (SPX), European options on the Dow Jones Industrial Average (DJX), and European options on the Nasdaq 100 (NDX). In Chapter 9, we explained that the CBOE trades LEAPS and flex options on individual stocks. It also offers these option products on indices.

One index option contract is on 100 times the index. (Note that the Dow Jones index used for index options is 0.01 times the usually quoted Dow Jones index.) Index options are settled in cash. This means that, on exercise of the option, the holder of a call option contract receives $(S - K) \times 100$ in cash and the writer of the option pays this amount in cash, where S is the value of the index at the close of trading on the day of the exercise and K is the strike price. Similarly, the holder of a put option contract receives $(K - S) \times 100$ in cash and the writer of the option pays this amount in cash.

Portfolio Insurance

Portfolio managers can use index options to limit their downside risk. Suppose that the value of an index today is S_0 . Consider a manager in charge of a well-diversified portfolio whose beta is 1.0. A beta of 1.0 implies that the returns from the portfolio mirror those

from the index. Assuming the dividend yield from the portfolio is the same as the dividend yield from the index, the percentage changes in the value of the portfolio can be expected to be approximately the same as the percentage changes in the value of the index. Since each contract is on 100 times the index, it follows that the value of the portfolio is protected against the possibility of the index falling below K if, for each $100S_0$ dollars in the portfolio, the manager buys one put option contract with strike price K . Suppose that the manager's portfolio is worth \$500,000 and the value of the index is 1,000. The portfolio is worth 500 times the index. The manager can obtain insurance against the value of the portfolio dropping below \$450,000 in the next three months by buying five three-month put option contracts on the index with a strike price of 900.

To illustrate how the insurance works, consider the situation where the index drops to 880 in three months. The portfolio will be worth about \$440,000. The payoff from the options will be $5 \times (900 - 880) \times 100 = \$10,000$, bringing the total value of the portfolio up to the insured value of \$450,000.

When the Portfolio's Beta Is Not 1.0

If the portfolio's beta (β) is not 1.0, β put options must be purchased for each $100S_0$ dollars in the portfolio, where S_0 is the current value of the index. Suppose that the \$500,000 portfolio just considered has a beta of 2.0 instead of 1.0. We continue to assume that the index is 1,000. The number of put options required is

$$2.0 \times \frac{500,000}{1,000 \times 100} = 10$$

rather than 5 as before.

To calculate the appropriate strike price, the capital asset pricing model can be used (see the appendix to Chapter 3). Suppose that the risk free rate is 12%, the dividend yield on both the index and the portfolio is 4%, and protection is required against the value of the portfolio dropping below \$450,000 in the next three months. Under the capital asset pricing model, the expected excess return of a portfolio over the risk-free

Table 16.1 Calculation of expected value of portfolio when the index is 1,040 in three months and $\beta = 2.0$.

Value of index in three months:	1,040
Return from change in index:	40/1,000, or 4% per three months
Dividends from index:	$0.25 \times 4 = 1\%$ per three months
Total return from index:	$4 + 1 = 5\%$ per three months
Risk-free interest rate:	$0.25 \times 12 = 3\%$ per three months
Excess return from index over risk-free interest rate:	$5 - 3 = 2\%$ per three months
Expected excess return from portfolio over risk-free interest rate:	$2 \times 2 = 4\%$ per three months
Expected return from portfolio:	$3 + 4 = 7\%$ per three months
Dividends from portfolio:	$0.25 \times 4 = 1\%$ per three months
Expected increase in value of portfolio:	$7 - 1 = 6\%$ per three months
Expected value of portfolio:	$\$500,000 \times 1.06 = \$530,000$

Table 16.2 Relationship between value of index and value of portfolio for $\beta = 2.0$.

<i>Value of index in three months</i>	<i>Value of portfolio in three months (\$)</i>
1,080	570,000
1,040	530,000
1,000	490,000
960	450,000
920	410,000
880	370,000

rate is assumed to equal beta times the excess return of the index portfolio over the risk-free rate. The model enables the expected value of the portfolio to be calculated for different values of the index at the end of three months. Table 16.1 shows the calculations for the case where the index is 1,040. In this case the expected value of the portfolio at the end of the three months is \$530,000. Similar calculations can be carried out for other values of the index at the end of the three months. The results are shown in Table 16.2. The strike price for the options that are purchased should be the index level corresponding to the protection level required on the portfolio. In this case the protection level is \$450,000 and so the correct strike price for the 10 put option contracts that are purchased is 960.¹

To illustrate how the insurance works, consider what happens if the value of the index falls to 880. As shown in Table 16.2, the value of the portfolio is then about \$370,000. The put options pay off $(960 - 880) \times 10 \times 100 = \$80,000$, and this is exactly what is necessary to move the total value of the portfolio manager's position up from \$370,000 to the required level of \$450,000.

The examples in this section show that there are two reasons why the cost of hedging increases as the beta of a portfolio increases. More put options are required and they have a higher strike price.

16.2 CURRENCY OPTIONS

Currency options are primarily traded in the over-the-counter market. The advantage of this market is that large trades are possible, with strike prices, expiration dates, and other features tailored to meet the needs of corporate treasurers. Although currency options do trade on NASDAQ OMX in the United States, the exchange-traded market for these options is much smaller than the over-the-counter market.

An example of a European call option is a contract that gives the holder the right to buy one million euros with US dollars at an exchange rate of 1.2000 US dollars per euro. If the actual exchange rate at the maturity of the option is 1.2500, the payoff is

¹ Approximately 1% of \$500,000, or \$5,000, will be earned in dividends over the next three months. If we want the insured level of \$450,000 to include dividends, we can choose a strike price corresponding to \$445,000 rather than \$450,000. This is 955.

$1,000,000 \times (1.2500 - 1.2000) = \$50,000$. Similarly, an example of a European put option is a contract that gives the holder the right to sell ten million Australian dollars for US dollars at an exchange rate of 0.9000 US dollars per Australian dollar. If the actual exchange rate at the maturity of the option is 0.8700, the payoff is $10,000,000 \times (0.9000 - 0.8700) = \$300,000$.

For a corporation wishing to hedge a foreign exchange exposure, foreign currency options are an alternative to forward contracts. A company due to receive sterling at a known time in the future can hedge its risk by buying put options on sterling that mature at that time. The hedging strategy guarantees that the exchange rate applicable to the sterling will not be less than the strike price, while allowing the company to benefit from any favorable exchange-rate movements. Similarly, a company due to pay sterling at a known time in the future can hedge by buying calls on sterling that mature at that time. This hedging strategy guarantees that the cost of the sterling will not be greater than a certain amount while allowing the company to benefit from favorable exchange-rate movements. Whereas a forward contract locks in the exchange rate for a future transaction, an option provides a type of insurance. This is not free. It costs nothing to enter into a forward transaction, but options require a premium to be paid up front.

Range Forwards

A *range forward contract* is a variation on a standard forward contract for hedging foreign exchange risk. Consider a US company that knows it will receive one million pounds sterling in three months. Suppose that the three-month forward exchange rate is 1.5200 dollars per pound. The company could lock in this exchange rate for the dollars it receives by entering into a short forward contract to sell one million pounds sterling in three months. This would ensure that the amount received for the one million pounds is \$1,520,000.

An alternative is to buy a European put option with a strike price of K_1 and sell a European call option with a strike price K_2 , where $K_1 < 1.5200 < K_2$. This is known as a short range forward contract. The payoff is shown in Figure 16.1a. In both cases, the options are on one million pounds. If the exchange rate in three months proves to be less than K_1 , the put option is exercised and as a result the company is able to sell the one

Figure 16.1 Payoffs from (a) short and (b) long range-forward contract.

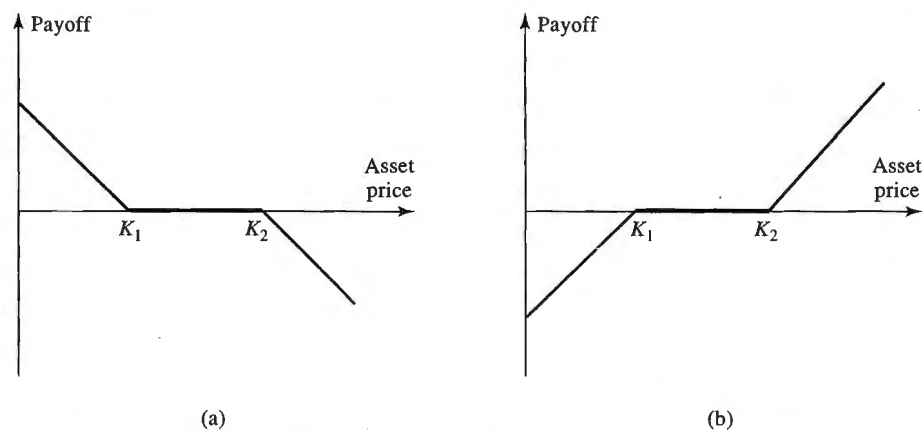
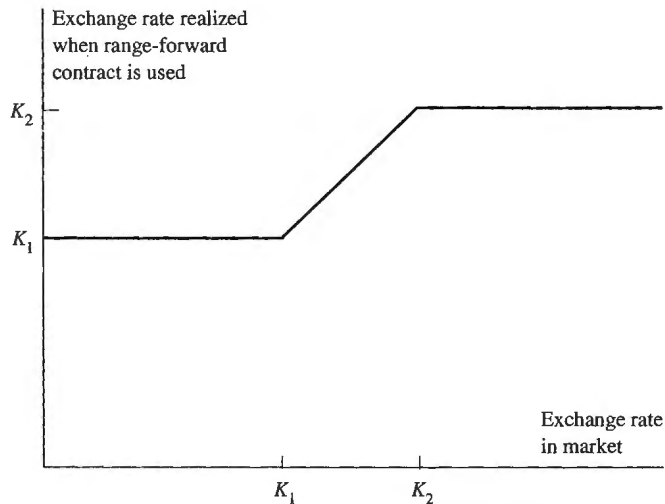


Figure 16.2 Exchange rate realized when either (a) a short range-forward contract is used to hedge a future foreign currency inflow or (b) a long range-forward contract is used to hedge a future foreign currency outflow.



million pounds at an exchange rate of K_1 . If the exchange rate is between K_1 and K_2 , neither option is exercised and the company gets the current exchange rate for the one million pounds. If the exchange rate is greater than K_2 , the call option is exercised against the company with the result that the one million pounds is sold at an exchange rate of K_2 . The exchange rate realized for the one million pounds is shown in Figure 16.2.

If the company knew it was due to pay rather than receive one million pounds in three months, it could sell a European put option with strike price K_1 and buy a European call option with strike price K_2 . This is known as a long range forward contract and the payoff is shown in Figure 16.1b. If the exchange rate in three months proves to be less than K_1 , the put option is exercised against the company and as a result the company buys the one million pounds it needs at an exchange rate of K_1 . If the exchange rate is between K_1 and K_2 , neither option is exercised and the company buys the one million pounds at the current exchange rate. If the exchange rate is greater than K_2 , the call option is exercised and the company is able to buy the one million pounds at an exchange rate of K_2 . The exchange rate paid for the one million pounds is the same as that received for the one million pounds in the earlier example and is shown in Figure 16.2.

In practice, a range forward contract is set up so that the price of the put option equals the price of the call option. This means that it costs nothing to set up the range forward contract, just as it costs nothing to set up a regular forward contract. Suppose that the US and British interest rates are both 5%, so that the spot exchange rate is 1.5200 (the same as the forward exchange rate). Suppose further that the exchange rate volatility is 14%. We can use DerivaGem to show that a European put with strike price 1.5000 to sell one pound has the same price as a European call option with a strike price of 1.5413 to buy one pound. (Both are worth 0.03250.) Setting $K_1 = 1.5000$ and $K_2 = 1.5413$ therefore leads to a contract with zero cost in our example.

In the limit, as the strike prices of the call and put options in a range forward contract are moved closer, the range forward contract becomes a regular forward contract. A short range forward contract becomes a short forward contract and a long range forward contract becomes a long forward contract.

16.3 OPTIONS ON STOCKS PAYING KNOWN DIVIDEND YIELDS

In this section we produce a simple rule that enables valuation results for European options on a non-dividend-paying stock to be extended so that they apply to European options on a stock paying a known dividend yield. Later we show how this enables us to value options on stock indices and currencies.

Dividends cause stock prices to reduce on the ex-dividend date by the amount of the dividend payment. The payment of a dividend yield at rate q therefore causes the growth rate in the stock price to be less than it would otherwise be by an amount q . If, with a dividend yield of q , the stock price grows from S_0 today to S_T at time T , then in the absence of dividends it would grow from S_0 today to $S_T e^{qT}$ at time T . Alternatively, in the absence of dividends it would grow from $S_0 e^{-qT}$ today to S_T at time T .

This argument shows that we get the same probability distribution for the stock price at time T in each of the following two cases:

1. The stock starts at price S_0 and provides a dividend yield at rate q .
2. The stock starts at price $S_0 e^{-qT}$ and pays no dividends.

This leads to a simple rule. When valuing a European option lasting for time T on a stock paying a known dividend yield at rate q , we reduce the current stock price from S_0 to $S_0 e^{-qT}$ and then value the option as though the stock pays no dividends.²

Lower Bounds for Option Prices

As a first application of this rule, consider the problem of determining bounds for the price of a European option on a stock paying a dividend yield at rate q . Substituting $S_0 e^{-qT}$ for S_0 in equation (10.4), we see that a lower bound for the European call option price, c , is given by

$$c \geq \max(S_0 e^{-qT} - Ke^{-rT}, 0) \quad (16.1)$$

We can also prove this directly by considering the following two portfolios:

Portfolio A: one European call option plus an amount of cash equal to Ke^{-rT}

Portfolio B: e^{-qT} shares with dividends being reinvested in additional shares.

To obtain a lower bound for a European put option, we can similarly replace S_0 by $S_0 e^{-qT}$ in equation (10.5) to get

$$p \geq \max(Ke^{-rT} - S_0 e^{-qT}, 0) \quad (16.2)$$

² This rule is analogous to the one developed in Section 14.12 for valuing a European option on a stock paying known cash dividends. (In that case we concluded that it is correct to reduce the stock price by the present value of the dividends; in this case we discount the stock price at the dividend yield rate.)

This result can also be proved directly by considering the following portfolios:

Portfolio C: one European put option plus e^{-qT} shares with dividends on the shares being reinvested in additional shares

Portfolio D: an amount of cash equal to Ke^{-rT} .

Put-Call Parity

Replacing S_0 by S_0e^{-qT} in equation (10.6) we obtain put-call parity for an option on a stock paying a dividend yield at rate q :

$$c + Ke^{-rT} = p + S_0e^{-qT} \quad (16.3)$$

This result can also be proved directly by considering the following two portfolios:

Portfolio A: one European call option plus an amount of cash equal to Ke^{-rT}

Portfolio C: one European put option plus e^{-qT} shares with dividends on the shares being reinvested in additional shares.

Both portfolios are both worth $\max(S_T, K)$ at time T . They must therefore be worth the same today, and the put-call parity result in equation (16.3) follows. For American options, the put-call parity relationship is (see Problem 16.12)

$$S_0e^{-qT} - K \leq C - P \leq S_0 - Ke^{-rT}$$

Pricing Formulas

By replacing S_0 by S_0e^{-qT} in the Black-Scholes-Merton formulas, equations (14.20) and (14.21), we obtain the price, c , of a European call and the price, p , of a European put on a stock paying a dividend yield at rate q as

$$c = S_0e^{-qT}N(d_1) - Ke^{-rT}N(d_2) \quad (16.4)$$

$$p = Ke^{-rT}N(-d_2) - S_0e^{-qT}N(-d_1) \quad (16.5)$$

Since

$$\ln \frac{S_0e^{-qT}}{K} = \ln \frac{S_0}{K} - qT$$

it follows that d_1 and d_2 are given by

$$d_1 = \frac{\ln(S_0/K) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/K) + (r - q - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

These results were first derived by Merton.³ As discussed in Chapter 14, the word *dividend* should, for the purposes of option valuation, be defined as the reduction in the stock price on the ex-dividend date arising from any dividends declared. If the dividend yield rate is known but not constant during the life of the option, equations (16.4)

³ See R. C. Merton, "Theory of Rational Option Pricing," *Bell Journal of Economics and Management Science*, 4 (Spring 1973): 141-83.

and (16.5) are still true, with q equal to the average annualized dividend yield during the option's life.

Differential Equation and Risk-Neutral Valuation

To prove the results in equations (16.4) and (16.5) more formally, we can either solve the differential equation that the option price must satisfy or use risk-neutral valuation.

When we include a dividend yield of q in the analysis in Section 14.6, the differential equation (14.16) becomes⁴

$$\frac{\partial f}{\partial t} + (r - q)S \frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} = rf \quad (16.6)$$

Like equation (14.16), this does not involve any variable affected by risk preferences. Therefore the risk-neutral valuation procedure described in Section 14.7 can be used.

In a risk-neutral world, the total return from the stock must be r . The dividends provide a return of q . The expected growth rate in the stock price must therefore be $r - q$. It follows that the risk-neutral process for the stock price is

$$dS = (r - q)S dt + \sigma S dz \quad (16.7)$$

To value a derivative dependent on a stock that provides a dividend yield equal to q , we set the expected growth rate of the stock equal to $r - q$ and discount the expected payoff at rate r . When the expected growth rate in the stock price is $r - q$, the expected stock price at time T is $S_0 e^{(r-q)T}$. A similar analysis to that in the appendix to Chapter 14 gives the expected payoff for a call option in a risk-neutral world as

$$e^{(r-q)T} S_0 N(d_1) - KN(d_2)$$

where d_1 and d_2 are defined as above. Discounting at rate r for time T leads to equation (16.4).

16.4 VALUATION OF EUROPEAN STOCK INDEX OPTIONS

In valuing index futures in Chapter 5, we assumed that the index could be treated as an asset paying a known yield. In valuing index options, we make similar assumptions. This means that equations (16.1) and (16.2) provide a lower bound for European index options; equation (16.3) is the put-call parity result for European index options; equations (16.4) and (16.5) can be used to value European options on an index; and the binomial tree approach can be used for American options. In all cases, S_0 is equal to the value of the index, σ is equal to the volatility of the index, and q is equal to the average annualized dividend yield on the index during the life of the option.

Example 16.1

Consider a European call option on the S&P 500 that is two months from maturity. The current value of the index is 930, the exercise price is 900, the risk-free interest rate is 8% per annum, and the volatility of the index is 20% per annum. Dividend

⁴ See Technical Note 6 at www.rotman.utoronto.ca/~hull/TechnicalNotes for a proof of this.

yields of 0.2% and 0.3% are expected in the first month and the second month, respectively. In this case $S_0 = 930$, $K = 900$, $r = 0.08$, $\sigma = 0.2$, and $T = 2/12$. The total dividend yield during the option's life is $0.2\% + 0.3\% = 0.5\%$. This corresponds to 3% per annum. Hence, $q = 0.03$ and

$$d_1 = \frac{\ln(930/900) + (0.08 - 0.03 + 0.2^2/2) \times 2/12}{0.2\sqrt{2/12}} = 0.5444$$

$$d_2 = \frac{\ln(930/900) + (0.08 - 0.03 - 0.2^2/2) \times 2/12}{0.2\sqrt{2/12}} = 0.4628$$

$$N(d_1) = 0.7069, \quad N(d_2) = 0.6782$$

so that the call price, c , is given by equation (16.4) as

$$c = 930 \times 0.7069e^{-0.03 \times 2/12} - 900 \times 0.6782e^{-0.08 \times 2/12} = 51.83$$

One contract would cost \$5,183.

The calculation of q should include only dividends for which the ex-dividend dates occur during the life of the option. In the United States ex-dividend dates tend to occur during the first week of February, May, August, and November. At any given time the correct value of q is therefore likely to depend on the life of the option. This is even more true for indices in other countries. In Japan, for example, all companies tend to use the same ex-dividend dates.

If the absolute amount of the dividend that will be paid on the stocks underlying the index (rather than the dividend yield) is assumed to be known, the basic Black-Scholes-Merton formulas can be used with the initial stock price being reduced by the present value of the dividends. This is the approach recommended in Chapter 14 for a stock paying known dividends. However, it may be difficult to implement for a broadly based stock index because it requires a knowledge of the dividends expected on every stock underlying the index.

It is sometimes argued that, in the long run, the return from investing a certain amount of money in a well-diversified stock portfolio is almost certain to beat the return from investing the same amount of money in a bond portfolio. If this were so, a long-dated put option allowing the stock portfolio to be sold for the value of the bond portfolio should not cost very much. In fact, as indicated by Business Snapshot 16.1, it is quite expensive.

Forward Prices

Define F_0 as the forward price of the index for a contract with maturity T . As shown by equation (5.3), $F_0 = S_0e^{(r-q)T}$. This means that the equations for the European call price c and the European put price p in equations (16.4) and (16.5) can be written

$$c = F_0e^{-rT}N(d_1) - Ke^{-rT}N(d_2) \quad (16.8)$$

$$p = Ke^{-rT}N(-d_2) - F_0e^{-rT}N(-d_1) \quad (16.9)$$

where

$$d_1 = \frac{\ln(F_0/K) + \sigma^2 T/2}{\sigma\sqrt{T}} \quad \text{and} \quad d_2 = \frac{\ln(F_0/K) - \sigma^2 T/2}{\sigma\sqrt{T}}$$

Business Snapshot 16.1 Can We Guarantee that Stocks Will Beat Bonds in the Long Run?

It is often said that if you are a long-term investor you should buy stocks rather than bonds. Consider a US fund manager who is trying to persuade investors to buy, as a long-term investment, an equity fund that is expected to mirror the S&P 500. The manager might be tempted to offer purchasers of the fund a guarantee that their return will be at least as good as the return on risk-free bonds over the next 10 years. Historically stocks have outperformed bonds in the United States over almost any 10-year period. It appears that the fund manager would not be giving much away.

In fact, this type of guarantee is surprisingly expensive. Suppose that an equity index is 1,000 today, the dividend yield on the index is 1% per annum, the volatility of the index is 15% per annum, and the 10-year risk-free rate is 5% per annum. To outperform bonds, the stocks underlying the index must earn more than 5% per annum. The dividend yield will provide 1% per annum. The capital gains on the stocks must therefore provide 4% per annum. This means that we require the index level to be at least $1,000e^{0.04 \times 10} = 1,492$ in 10 years.

A guarantee that the return on \$1,000 invested in the index will be greater than the return on \$1,000 invested in bonds over the next 10 years is therefore equivalent to the right to sell the index for 1,492 in 10 years. This is a European put option on the index and can be valued from equation (16.5) with $S_0 = 1,000$, $K = 1,492$, $r = 5\%$, $\sigma = 15\%$, $T = 10$, and $q = 1\%$. The value of the put option is 169.7. This shows that the guarantee contemplated by the fund manager is worth about 17% of the fund—hardly something that should be given away!

The put-call parity relationship in equation (16.3) can be written

$$c + Ke^{-rT} = p + F_0e^{-rT}$$

or

$$F_0 = K + (c - p)e^{rT} \quad (16.10)$$

If, as is not uncommon in the exchange-traded markets, pairs of puts and calls with the same strike price are traded actively for a particular maturity date, this equation can be used to estimate the forward price of the index for that maturity date. Once the forward prices of the index for a number of different maturity dates have been obtained, the term structure of forward prices can be estimated, and other options can be valued using equations (16.8) and (16.9). The advantage of this approach is that the dividend yield on the index does not have to be estimated explicitly.

Implied Dividend Yields

If estimates of the dividend yield are required (e.g., because an American option is being valued), calls and puts with the same strike price and time to maturity can again be used. From equation (16.3),

$$q = -\frac{1}{T} \ln \frac{c - p + Ke^{-rT}}{S_0}$$

For a particular strike price and time to maturity, the estimates of q calculated from this equation are liable to be unreliable. But when the results from many matched pairs of calls and puts are combined, a clearer picture of the dividend yield being assumed by the market emerges.

16.5 VALUATION OF EUROPEAN CURRENCY OPTIONS

To value currency options, we define S_0 as the spot exchange rate. To be precise, S_0 is the value of one unit of the foreign currency in US dollars. As explained in Section 5.10, a foreign currency is analogous to a stock paying a known dividend yield. The owner of foreign currency receives a yield equal to the risk-free interest rate, r_f , in the foreign currency. Equations (16.1) and (16.2), with q replaced by r_f , provide bounds for the European call price, c , and the European put price, p :

$$c \geq \max(S_0 e^{-r_f T} - K e^{-r T}, 0)$$

$$p \geq \max(K e^{-r T} - S_0 e^{-r_f T}, 0)$$

Equation (16.3), with q replaced by r_f , provides the put-call parity result for European currency options:

$$c + K e^{-r T} = p + S_0 e^{-r_f T}$$

Finally, equations (16.4) and (16.5) provide the pricing formulas for European currency options when q is replaced by r_f :

$$c = S_0 e^{-r_f T} N(d_1) - K e^{-r T} N(d_2) \quad (16.11)$$

$$p = K e^{-r T} N(-d_2) - S_0 e^{-r_f T} N(-d_1) \quad (16.12)$$

where

$$d_1 = \frac{\ln(S_0/K) + (r - r_f + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/K) + (r - r_f - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

Both the domestic interest rate, r , and the foreign interest rate, r_f , are the rates for a maturity T . Put and call options on a currency are symmetrical in that a put option to sell currency A for currency B at strike price K is the same as a call option to buy B with currency A at strike price $1/K$ (see Problem 16.8).

Example 16.2

Consider a four-month European call option on the British pound. Suppose that the current exchange rate is 1.6000, the exercise price is 1.6000, the risk-free interest rate in the United States is 8% per annum, the risk-free interest rate in Britain is 11% per annum, and the option price is 4.3 cents. In this case, $S_0 = 1.6$, $K = 1.6$, $r = 0.08$, $r_f = 0.11$, $T = 0.3333$, and $c = 0.043$. The implied volatility can be calculated by trial and error. A volatility of 20% gives an option price of 0.0639; a volatility of 10% gives an option price of 0.0285; and so on. The implied volatility is 14.1%.

Using Forward Exchange Rates

Because banks and other financial institutions trade forward contracts on foreign exchange rates actively, foreign exchange rates are often used for valuing options.

From equation (5.9), the forward rate, F_0 , for a maturity T is given by

$$F_0 = S_0 e^{(r-r_f)T}$$

This relationship allows equations (16.11) and (16.12) to be simplified to

$$c = e^{-rT} [F_0 N(d_1) - KN(d_2)] \quad (16.13)$$

$$p = e^{-rT} [KN(-d_2) - F_0 N(-d_1)] \quad (16.14)$$

where

$$d_1 = \frac{\ln(F_0/K) + \sigma^2 T/2}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(F_0/K) - \sigma^2 T/2}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

Equations (16.13) and (16.14) are the same as equations (16.8) and (16.9). As we shall see in Chapter 17, a European option on the spot price of any asset can be valued in terms of the price of a forward or futures contract on the asset using equations (16.13) and (16.14). The maturity of the forward or futures contract must be the same as the maturity of the European option.

16.6 AMERICAN OPTIONS

As described in Chapter 12, binomial trees can be used to value American options on indices and currencies. As in the case of American options on a non-dividend-paying stock, the parameter determining the size of up movements, u , is set equal to $e^{\sigma\sqrt{\Delta t}}$, where σ is the volatility and Δt is the length of time steps. The parameter determining the size of down movements, d , is set equal to $1/u$, or $e^{-\sigma\sqrt{\Delta t}}$. For a non-dividend-paying stock, the probability of an up movement is

$$p = \frac{a - d}{u - d}$$

where $a = e^{r\Delta t}$. For options on indices and currencies, the formula for p is the same, but a is defined differently. In the case of options on an index,

$$a = e^{(r-q)\Delta t} \quad (16.15)$$

where q is the dividend yield on the index. In the case of options on a currency,

$$a = e^{(r-r_f)\Delta t} \quad (16.16)$$

where r_f is the foreign risk-free rate. Example 12.1 in Section 12.11 shows how a two-step tree can be constructed to value an option on an index. Example 12.2 shows how a three-step tree can be constructed to value an option on a currency. Further examples of the use of binomial trees to value options on indices and currencies are given in Chapter 20.

In some circumstances, it is optimal to exercise American currency and index options prior to maturity. Thus, American currency and index options are worth more than their European counterparts. In general, call options on high-interest currencies and put options on low-interest currencies are the most likely to be exercised prior to maturity. The reason is that a high-interest currency is expected to depreciate and a low-interest currency is expected to appreciate. In addition, call options on indices with high-dividend yields and put options on indices with low-dividend yields are most likely to be exercised early.

SUMMARY

The index options that trade on exchanges are settled in cash. On exercise of an index call option, the holder receives 100 times the amount by which the index exceeds the strike price. Similarly, on exercise of an index put option contract, the holder receives 100 times the amount by which the strike price exceeds the index. Index options can be used for portfolio insurance. If the value of the portfolio mirrors the index, it is appropriate to buy one put option contract for each $100S_0$ dollars in the portfolio, where S_0 is the value of the index. If the portfolio does not mirror the index, β put option contracts should be purchased for each $100S_0$ dollars in the portfolio, where β is the beta of the portfolio calculated using the capital asset pricing model. The strike price of the put options purchased should reflect the level of insurance required.

Most currency options are traded in the over-the-counter market. They can be used by corporate treasurers to hedge a foreign exchange exposure. For example, a US corporate treasurer who knows that the company will be receiving sterling at a certain time in the future can hedge by buying put options that mature at that time. Similarly, a US corporate treasurer who knows that the company will be paying sterling at a certain time in the future can hedge by buying call options that mature at that time. Currency options can also be used to create a range forward contract. This is a zero-cost contract that can be used to provide downside protection while giving up some of the upside for a company with a known foreign exchange exposure.

The Black-Scholes-Merton formula for valuing European options on a non-dividend-paying stock can be extended to cover European options on a stock paying a known dividend yield. The extension can be used to value European options on stock indices and currencies because:

1. A stock index is analogous to a stock paying a dividend yield. The dividend yield is the dividend yield on the stocks that make up the index.
2. A foreign currency is analogous to a stock paying a dividend yield. The foreign risk-free interest rate plays the role of the dividend yield.

Binomial trees can be used to value American options on stock indices and currencies.

FURTHER READING

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Practice Questions (Answers in Solutions Manual)

- 16.1. A portfolio is currently worth \$10 million and has a beta of 1.0. An index is currently standing at 800. Explain how a put option on the index with a strike price of 700 can be used to provide portfolio insurance.
- 16.2. "Once we know how to value options on a stock paying a dividend yield, we know how to value options on stock indices and currencies." Explain this statement.
- 16.3. A stock index is currently 300, the dividend yield on the index is 3% per annum, and the risk-free interest rate is 8% per annum. What is a lower bound for the price of a six-month European call option on the index when the strike price is 290?
- 16.4. A currency is currently worth \$0.80 and has a volatility of 12%. The domestic and foreign risk-free interest rates are 6% and 8%, respectively. Use a two-step binomial tree to value (a) a European four-month call option with a strike price of 0.79 and (b) an American four-month call option with the same strike price.
- 16.5. Explain how corporations can use range forward contracts to hedge their foreign exchange risk when they are due to receive a certain amount of a foreign currency in the future.
- 16.6. Calculate the value of a three-month at-the-money European call option on a stock index when the index is at 250, the risk-free interest rate is 10% per annum, the volatility of the index is 18% per annum, and the dividend yield on the index is 3% per annum.
- 16.7. Calculate the value of an eight-month European put option on a currency with a strike price of 0.50. The current exchange rate is 0.52, the volatility of the exchange rate is 12%, the domestic risk-free interest rate is 4% per annum, and the foreign risk-free interest rate is 8% per annum.
- 16.8. Show that the formula in equation (16.12) for a put option to sell one unit of currency A for currency B at strike price K gives the same value as equation (16.11) for a call option to buy K units of currency B for currency A at strike price $1/K$.
- 16.9. A foreign currency is currently worth \$1.50. The domestic and foreign risk-free interest rates are 5% and 9%, respectively. Calculate a lower bound for the value of a six-month call option on the currency with a strike price of \$1.40 if it is (a) European and (b) American.

- 16.10. Consider a stock index currently standing at 250. The dividend yield on the index is 4% per annum, and the risk-free rate is 6% per annum. A three-month European call option on the index with a strike price of 245 is currently worth \$10. What is the value of a three-month put option on the index with a strike price of 245?
- 16.11. An index currently stands at 696 and has a volatility of 30% per annum. The risk-free rate of interest is 7% per annum and the index provides a dividend yield of 4% per annum. Calculate the value of a three-month European put with an exercise price of 700.
- 16.12. Show that, if C is the price of an American call with exercise price K and maturity T on a stock paying a dividend yield of q , and P is the price of an American put on the same stock with the same strike price and exercise date, then

$$S_0 e^{-qT} - K < C - P < S_0 - K e^{-rT},$$

where S_0 is the stock price, r is the risk-free rate, and $r > 0$. (*Hint*: To obtain the first half of the inequality, consider possible values of:

Portfolio A: a European call option plus an amount K invested at the risk-free rate

Portfolio B: an American put option plus e^{-qT} of stock with dividends being reinvested in the stock.

To obtain the second half of the inequality, consider possible values of:

Portfolio C: an American call option plus an amount $K e^{-rT}$ invested at the risk-free rate

Portfolio D: a European put option plus one stock with dividends being reinvested in the stock.)

- 16.13. Show that a European call option on a currency has the same price as the corresponding European put option on the currency when the forward price equals the strike price.
- 16.14. Would you expect the volatility of a stock index to be greater or less than the volatility of a typical stock? Explain your answer.
- 16.15. Does the cost of portfolio insurance increase or decrease as the beta of a portfolio increases? Explain your answer.
- 16.16. Suppose that a portfolio is worth \$60 million and the S&P 500 is at 1,200. If the value of the portfolio mirrors the value of the index, what options should be purchased to provide protection against the value of the portfolio falling below \$54 million in one year's time?
- 16.17. Consider again the situation in Problem 16.16. Suppose that the portfolio has a beta of 2.0, the risk-free interest rate is 5% per annum, and the dividend yield on both the portfolio and the index is 3% per annum. What options should be purchased to provide protection against the value of the portfolio falling below \$54 million in one year's time?
- 16.18. An index currently stands at 1,500. European call and put options with a strike price of 1,400 and time to maturity of six months have market prices of 154.00 and 34.25, respectively. The six-month risk-free rate is 5%. What is the implied dividend yield?
- 16.19. A total return index tracks the return, including dividends, on a certain portfolio. Explain how you would value (a) forward contracts and (b) European options on the index.
- 16.20. What is the put-call parity relationship for European currency options?

- 16.21. Can an option on the yen–euro exchange rate be created from two options, one on the dollar–euro exchange rate, and the other on the dollar–yen exchange rate? Explain your answer.
- 16.22. Prove the results in equations (16.1), (16.2), and (16.3) using the portfolios indicated.

Further Questions

- 16.23. The Dow Jones Industrial Average on January 12, 2007, was 12,556 and the price of the March 126 call was \$2.25. Use the DerivaGem software to calculate the implied volatility of this option. Assume the risk-free rate was 5.3% and the dividend yield was 3%. The option expires on March 20, 2007. Estimate the price of a March 126 put. What is the volatility implied by the price you estimate for this option? (Note that options are on the Dow Jones index divided by 100.)
- 16.24. A stock index currently stands at 300 and has a volatility of 20%. The risk-free interest rate is 8% and the dividend yield on the index is 3%. Use a three-step binomial tree to value a six-month put option on the index with a strike price of 300 if it is (a) European and (b) American?
- 16.25. Suppose that the spot price of the Canadian dollar is US \$0.95 and that the Canadian dollar/US dollar exchange rate has a volatility of 8% per annum. The risk-free rates of interest in Canada and the United States are 4% and 5% per annum, respectively. Calculate the value of a European call option to buy one Canadian dollar for US \$0.95 in nine months. Use put–call parity to calculate the price of a European put option to sell one Canadian dollar for US \$0.95 in nine months. What is the price of a call option to buy US \$0.95 with one Canadian dollar in nine months?
- 16.26. Hedge funds earn a fixed fee plus a percentage of the profits, if any, that they generate (see Business Snapshot 1.2). How is a fund manager motivated to behave with this type of arrangement?
- 16.27. Assume that the price of currency A expressed in terms of the price of currency B follows the process $dS = (r_B - r_A)S dt + \sigma S dz$, where r_A is the risk-free interest rate in currency A and r_B is the risk-free interest rate in currency B. What is the process followed by the price of currency B expressed in terms of currency A?
- 16.28. The USD/euro exchange rate is 1.3000. The exchange rate volatility is 15%. A US company will receive 1 million euros in three months. The euro and USD risk-free rates are 5% and 4%, respectively. The company decides to use a range forward contract with the lower strike price equal to 1.2500.
- What should the higher strike price be to create a zero-cost contract?
 - What position in calls and puts should the company take?
 - Show that your answer to (a) does not depend on interest rates provided that the interest rate differential between the two currencies, $r - r_f$, remains the same.
- 16.29. In Business Snapshot 16.1, what is the cost of a guarantee that the return on the fund will not be negative over the next 10 years?